

[Fluence Guides](#)[Informational Videos](#)[FAQ](#)[Release Notes](#)

## Categories

[Fluence Technologies Help Center](#) / [Fluence Guides](#) / [Integration Toolkit](#)

# Fluence Rest API: Full Article

[Follow](#)

## About Fluence's REST API

The Fluence API allows integration developers to manage workflow and retrieve, insert, update, and delete financial data and metadata within their environments through external process automation. By leveraging the Fluence REST API, you can programmatically access, manipulate, and analyze financial data, automating your financial reporting and consolidation processes.

Developers have the ability to load and export data, retrieve a list of model dimensions along with their corresponding properties, as well as access lists of dimension members and their associated properties.

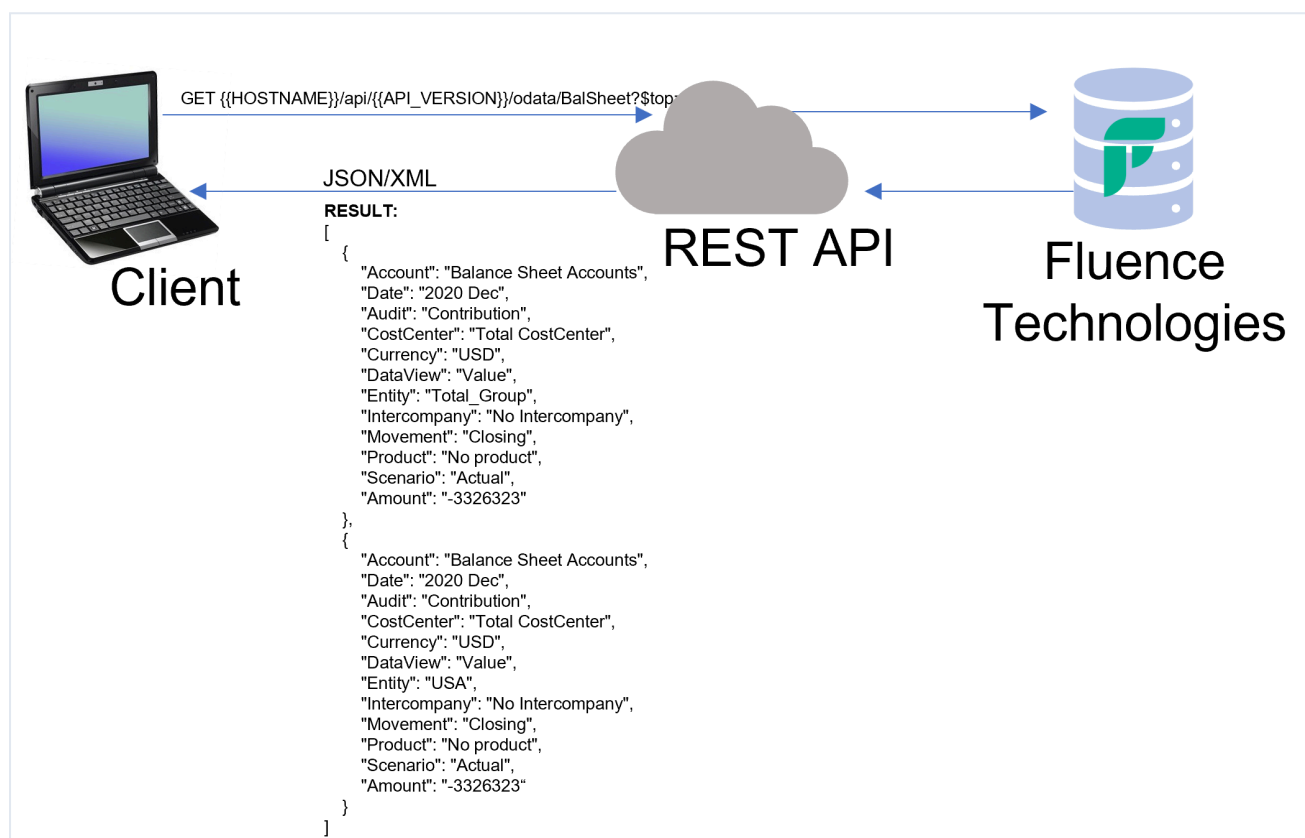
Developers can also initiate Fluence Workflow processes, monitor the status of ongoing workflows, and halt workflows if needed. This documentation provides an overview of the API authentication, endpoints, requests or responses, and parameters used.

Fluence's REST API is a secure, web-based Application Programming Interface that provides requests and responses between Fluence and other cloud-based systems. Once a connection string is established by an admin in Fluence, and an encrypted API key is provided, the developer can make requests to the Fluence API service running on Microsoft Azure cloud. The Fluence engine processes any requests to the customer tenant database and Fluence responds with the requested information in JSON structured response format. The protocol used to transport the data is secure https. "REST" stands for Representational State Transfer, which is a stateless communication protocol for web services.

## Example

Some examples of use cases for the REST API against Fluence Consolidations would be retrieving Fluence financial Actual data to incorporate in a downstream Planning application or data lake; Updating Fluence data from an external source like an ERP system or from a middleware ETL tool; or pushing plan data into Fluence from a planning system to generate management budget reports.

In the example below, the GET method retrieves rows from a Fluence Consolidation Model for all tagged "Balance Sheet Accounts".



## Purpose of using Fluence's REST API

You can use Fluence's Rest API for any of the following:

1. **Endpoint Discovery:** Determine the REST API endpoints provided by Fluence.
2. **Retrieving Data:** Use the appropriate root URL, path parameters, and query parameters to retrieve data from Fluence. For example, you might fetch profit and loss, balance sheet, or cash flow data for specific periods or entities to view alongside your sales data for a particular entity and period within a planning solution.
3. **Synchronize Dimension Information:** Retrieve a list of dimensions, along with their parent-child relationships and other properties and attributes from Fluence, into a

planning solution to synchronized metadata and hierarchy information across both systems.

4. **Automation and Integration:** Use Fluence REST API calls to automate starting, stopping, and retrieving the Run ID of your workflows. For example, start a “Month-End Close” workflow to perform real-time processes and calculations to close the month-end for your organization from an external process manager, web page or automation manager.

## Authentication

Using the HTTPS protocol ensures that the client and server communication is encrypted and secure, providing an additional layer of protection for your REST API interactions.

To start, you will need the HOSTNAME, TENANT, X\_API\_TOKEN, and API VERSION for authentication for your Fluence environment.

### HOSTNAME

The name of the current REST API URL of your Fluence environment. The URL differs from your usual Fluence portal URL. To ensure Fluence's REST API works, you must utilize the URL below. If you use a development environment and would like to use the REST API against it, you only need to modify "prod" to "dev" in the example below.

**Example:** `https://fluenceapi-prod.fluence.app`

### TENANT

The name of the tenant of the Fluence Portal receiving requests and sending responses. The name of the tenant must be exact. If the tenant contains a space in the name, you must include the space, likewise for an underscore.

**Example:** Consolidation Prod

## API\_TOKEN

To obtain a token, an individual must be authorized to create user API Tokens. The generated token operates as the user for whom it was created. Users within the Fluence portal acquire tokens by following the steps outlined below:

1. Log in to the Fluence portal using administrator credentials.
2. Navigate to the Security module.
3. Click on the User API Token icon.
4. Click the New API Token icon.
5. In the Description box, type a Description, such as, "The import of planning data and the exportation of consolidated data to the planning solution will be facilitated through the utilization of the REST API."
6. For the Expiration Date, click the calendar icon, and choose the date the token will expire.
7. Click Save.

Users within the Fluence portal can also delete tokens they have created for themselves.

**NOTE:** Users of Fluence who have been granted access to the User API Tokens page can create and assign REST API tokens for themselves within the Fluence portal. It is imperative to ensure that the user acknowledges the receipt of the token and comprehends its purpose and guidelines for usage.

**Example:** 12345bc3-2929-599c-abc1-23f90aa94x3f

## API\_Version

The current API Version for the Fluence Portal is used in the URL request.

**Example:** v2305.1

**Authentication Example:****HOSTNAME:** https://fluenceapi-prod.fluence.app**TENANT:** Consolidation Prod**API\_TOKEN:** 12345bc3-2929-599c-abc1-23f90aa94x3f**API\_VERSION:** v2305.1

## The Anatomy of a Request

Anytime a call is made to a server using a REST API, that is considered an API integration request. The following is an example of a request:

**Example:****GET https://fluenceapi-prod.fluence.app//api/ v2305.1/odata/ClientSchemaTable****GET {{HOSTNAME}}/api/{{API\_VERSION}}/odata/ClientSchemaTable**

In the example above:

- where {{HOSTNAME}} is, it represents the base URL of the REST API:
  - **https://fluenceapi-prod.fluence.app/.**
- where {{API\_VERSION}} is, it represents the current API Version for the Fluence Portal used in the URL request:
  - **v2305.1**
- where ClientSchemaTable is, it represents the Fluence Client Database Schema Table:
  - **BalSheet**

## The Method

HTTP methods, which will be explained in further detail, are integral to a RESTful API request. These are the methods within ODATA of the Fluence REST API – GET, POST, PUT,

and DELETE. The START and STOP methods are also used to start and stop a Fluence workflow.

## HOSTNAME or ROOT URL

The `{{HOSTNAME}}` variable, also called a root URL of a REST API, is a unique URL representing an object or group of data objects. The Fluence REST API request has its own root URL, which the HTTP client is directed at to interact with data resources. The service root URL MUST terminate in a forward slash. Replace the variable "`{{HOSTNAME}}`" with the current REST API URL.

Example: GET `https://fluenceapi-prod.fluence.app/`

## EndPoint or Resource Path

A resource path refers to the part of the API's endpoint URL that identifies and locates a specific resource or collection of resources. It is typically a hierarchical representation of the resource's location within the API's data model.

A resource path comprises one or more path segments, separated by forward slashes ("/"). Each path segment may contain alphanumeric characters, as well as certain special characters like hyphens ("-") or underscores ("\_"). The path segments navigate through the API's resource hierarchy and identify the desired resource.

In the example below, the request's endpoint would be `/api/v2305.1/odata/`.

**Example:** GET `https://fluenceapi-prod.fluence.app/api/v2305.1/odata/`

## API

The field `"api/"` is mandatory in Fluence's REST API requests.

**Example:** GET `https://fluenceapi-prod.fluence.app/api/`

## API\_VERSION

The request needs the current API Version for the Fluence environment.

**Example:** GET https://fluenceapi-prod.fluence.app/api/v2305.1

## ODATA

ODATA (Open Data Protocol) is an OASIS standard defining best practices for building and consuming RESTful APIs. Fluence's REST API utilizes HTTP requests.

The field "odata/" is mandatory in Fluences's ODATA REST API requests. The ClientPathParam is the path parameter. Additionally, you can add optional query parameters after the Path Parameter.

**Example:** GET https://fluenceapi-prod.fluence.app/api/v2305.1/odata/ClientPathParam

## Client Schema Table or Path Parameter

The name of the client schema table within the named tenant of the Fluence Portal to receive requests and send responses. The client schema table is also known as a path parameter.

**Example:** GET https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation Prod

## Query Parameters

Query Parameters serve as criteria, for example, filtering, sorting, and allowing API requests to retrieve specific data. Use logical operators within query parameters.

API query parameters are optional key-value pairs that appear after the question mark in the URL. They serve as URL extensions and help determine specific content or actions based on the delivered data. To append query parameters, use a '?' at the end of the URL, followed by the parameter name and value.

Multiple query parameters can be added by separating them with an '&' sign, forming a query string. Query parameters can accommodate various object types with varying lengths, including strings and numbers.

The use of query parameters is warranted in several business cases, defined later in this documentation.

## The Headers

REST headers contain information representing the authentication token, tenant, and metadata associated with every REST API request. A REST header indicates the format of the request and response and provides information about the request's status.

### Example:

GET https://fluenceapi-prod.fluence.app/api/v2305.1/odata/**Consolidation Prod**

**Host:** fluenceapi-prod.fluence.app

**Authorization Bearer:** 12345bc3-2929-599c-abc1-23f90aa94x3f

**Content-Type:** application/json

In the example above, we have three headers:

1. **Host:** Specifies the server's domain name or IP address where the API is hosted. It helps the server identify which endpoint is being targeted.
2. **Authorization:** Provides authentication information for the request. This example uses the API Key authentication scheme with Key X-API-TOKEN, where the token value represents the API token.
3. **Content-Type:** Indicates the data format sent in the request body. This case specifies that the body content is in JSON format.

These headers provide essential information to the server, enabling it to process the request correctly and securely.

## The Body Data

A REST API request also consists of data (referred to as a "body") that usually works with the POST and PUT HTTP methods. The body data refers to the payload or content sent



along with the request. It contains the data that needs to be transmitted to the server for processing, such as JSON, or CVS data. The body data is typically included in the body of the HTTP request.

**Example:**

POST [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo)

**Body**

```
{
  "Account_Name": "6001",
  "Audit_Name": "Reclassification",
  "CostCenter_Name": "000",
  "Currency_Name": "USA",
  "DataView_Name": "Value",
  "Date_Name": "2023 Dec",
  "Entity_Name": "120",
  "Intercompany_Name": "No Intercompany ",
  "Movement_Name": "Closing",
  "Product_Name": "No product",
  "Scenario_Name": "Budget",
  "Amount": 25000.00,
  "Text": null,
  "Periodic_Amount": null,
}
```

In the example above, the request is a POST method to create a new row of budget data within the Consolidation\_Demo client schema table. The body data is represented in JSON format and includes information about the data. In this case, the body data reflects which member for the intersection of Account, Audit, CostCenter, Currency, DataView, Date, Entity, Intercompany, Movement, Product, Scenario, Measure (Amount, text, and Periodic\_Amount) will be used to create a new row in the table.

It's important to note that not all REST API requests require a body. For example, GET requests typically don't have a body since they retrieve data from the server rather than sending data to it. However, methods like POST and PUT often require a body to send data for creation or updating purposes.

## REST API Response

In a REST API request, the REST API response is the server's reply to the client's request. It contains the data or information requested by the client. The response is typically sent back

to the client in the body of an HTTP response. The response payload can be whatever is practical: Data responses are typically JSON.

**EXAMPLE:**

GET [https://fluenceapi-](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/PlanningActuals?$select=Account,Date,Entity,Scenario,Amount)

[prod.fluence.app/api/v2305.1/odata/PlanningActuals?\\$select=Account,Date,Entity,Scenario,Amount](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/PlanningActuals?$select=Account,Date,Entity,Scenario,Amount)

**Status: 200 OK**

**Content-Type: application/json**

**Body**

```
[
  {
    "Account": "1000",
    "Date": "2020 Feb",
    "Entity": "240",
    "Scenario": "Actual",
    "Amount": 519817.0
  },
  {
    "Account": "1000",
    "Date": "2020 Feb",
    "Entity": "260",
    "Scenario": "Actual",
    "Amount": 10679.0
  },
  {
    "Account": "1000",
    "Date": "2020 Jan",
    "Entity": "240",
    "Scenario": "Actual",
    "Amount": 535436.0
  },
]
```

In the example above, the response includes the following elements:

- **'Status'**: Indicates the status of the request. It can be "success" or "error".
- **'Code'**: Represents a numerical status code. In this case, it is 200 OK, indicating a successful request.
- **'Data'**: Contains the actual data returned by the API. In this case, the data within the fields Account, Date, Entity, Scenario, Amount.

## OData Overview

ODATA provides a best-practice approach to exposing and consuming data over the web, allowing clients to interact with data sources, such as Fluence environments, using a

consistent set of conventions and operations. It promotes interoperability between different systems and platforms by defining uniform operations and query options.

ODATA is designed to be simple, open, and widely supported. The ODATA methods can retrieve, insert, update, and delete data from a client staging table.

ODATA (Open Data Protocol) is an OASIS standard defining best practices for building and consuming RESTful APIs. Fluence's REST API utilizes HTTP requests.

The field "odata/" is mandatory in Fluences's ODATA REST API requests.

**Example:** GET <https://fluenceapi-prod.fluence.app/api/v2305.1/odata/ClientSchemaTable>

### GET Method

The GET request retrieves information. No matter how many times it repeats with identical parameters, the results will always be the same.

### POST Method

The POST method is used to insert data into the client schema table. Additionally, use the POST method to start and stop a workflow.

### PUT Method

The PUT method updates a row or rows within the client schema table.

### DELETE Method

The DELETE method removes a row or rows within a client schema table.

## Data types for parameters

The following are data types for parameters formatted in JSON.

These data types are the most common with REST APIs:

- **string:** An alphanumeric sequence of letters and numbers.

**Example:**

```
{
  "Scenario": "Actual"
}
```

- **integer:** A whole number — can be positive or negative  
Example

```
{
  "Amount": 519817.0
}
```

- **Boolean:** 1 for True or 0 for a false value

# GET Method

The GET request retrieves information from the Fluence server’s client schema table. By specifying the relevant parameters in the GET request, users can retrieve consolidated financial information such as balance sheet, profit and loss, cash flow data, and other financial metrics required for planning purposes, for example. This integration facilitates data sharing, synchronization, and real-time updates across different systems, enhancing overall operational efficiency and accuracy of financial information.

**Example of a GET for OData:**

The following GET method uses the Fluence endpoint and targets the BalSheet client schema table. It will only retrieve data in the schema table with the Date member names that start with 2020.

**EXAMPLE:**

```
GET https://fluenceapi-prod.fluence.app/api/v2305.1/odata/BalSheet/?$filter=startswith(Date,'2020')
```

| Option                               | Description                                                              |
|--------------------------------------|--------------------------------------------------------------------------|
| GET                                  | REST API Method being used. GET retrieves information.                   |
| https://fluenceapi-prod.fluence.app/ | This is the root URL of your Fluence environment's current REST API URL. |
| api/                                 | This endpoint indicates the usage of the API.                            |

|                                       |                                                                                                                                                                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v2305.1/                              | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                                                                                                                                                |
| Odata                                 | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API.                                                                                                                        |
| BalSheet                              | This path parameter is the name of the client schema table within your Fluence environment.<br><b>Example:</b> GET https://fluenceapi-prod.fluence.app/api/v2305.1/odata/ <b>BalSheet</b>                                                                     |
| ?<br>\$filter=startswith(Date,'2020') | The query parameter filters the Date dimension using a logical operator "=" to return any member name starting with '2020'.<br><b>Example:</b> GET https://fluenceapi-prod.fluence.app/api/v2305.1/odata/BalSheet?<br><b>\$filter=startswith(Date,'2020')</b> |
| Date                                  | This is a column name within the BalSheet client schema table.                                                                                                                                                                                                |
| '2020'                                | The field returned from the Date column of the BalSheet client schema table.                                                                                                                                                                                  |

### Results of the GET Method above:

Below is a sample of the results from the GET method above:

```
[
  {
    "Account": "Balance Sheet Accounts",
    "Date": "2020 Dec",
    "Audit": "Contribution",
    "CostCenter": "Total CostCenter",
    "Currency": "USD",
    "DataView": "Value",
```

```
"Entity": "Total_Group",

"Intercompany": "No Intercompany",

"Movement": "Closing",

"Product": "No product",

"Scenario": "Actual",

"Amount": "-3326323"

},

{

"Account": "Balance Sheet Accounts",

"Date": "2020 Dec",

"Audit": "Contribution",

"CostCenter": "Total CostCenter",

"Currency": "USD",

"DataView": "Value",

"Entity": "USA",

"Intercompany": "No Intercompany",

"Movement": "Closing",

"Product": "No product",

"Scenario": "Actual",

"Amount": "-3326323"

}

]
```

**NOTE:** More results were returned but are removed for brevity.

## POST Method

Use the POST method to insert financial data, such as budget or forecast data for a particular accounting period or entity, to compare your Actual results.

To utilize the POST method, it is essential to include specific components within the URL, namely the endpoint, API, version, OData, and client schema table name, similar to the GET method. These elements collectively form the foundation for the request.

Unlike the GET method, the data intended for insertion into the client schema table should be included within the request body, as in the example below.

**Example of inserting a single row:**

Utilize the POST method to insert a new row into Fluence’s client schema table with budget data from your planning solution, as in the example below.

POST https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\_Demo

Body

```
{
  "Account_Name": "6001",
  "Audit_Name": "Reclassification",
  "CostCenter_Name": "000",
  "Currency_Name": "USA",
  "DataView_Name": "Value",
  "Date_Name": "2023 Dec",
  "Entity_Name": "120",
  "Intercompany_Name": "No Intercompany ",
  "Movement_Name": "Closing",
  "Product_Name": "No product",
  "Scenario_Name": "Budget",
  "Amount": 25000.00,
  "Text": null,
  "Periodic_Amount": null,
}
```

| Option                               | Description                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| POST                                 | REST API Method being used. POST method inserts data into the client schema table.                             |
| https://fluenceapi-prod.fluence.app/ | This is the root URL of your Fluence environment's current REST API URL.                                       |
| api/                                 | This endpoint indicates the usage of the API.                                                                  |
| v2305.1/                             | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL. |

|                    |                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Odata              | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API. |
| Consolidation_Demo | The name of the client schema table within your Fluence environment.                                                                   |
| Body               | Inserts a new row within the client schema table based on the metadata intersection within the body.                                   |

**NOTE:** To determine if the POST method worked properly, utilize the GET method using the \$filter query method with the intersection of the metadata.

A batch insert refers to the capability of sending multiple data entries or objects in a single request to create them simultaneously on the server side. Rather than sending separate POST requests for each individual entry, a batch insert allows you to group them together and submit them as a batch.

Batch inserts can be beneficial in several ways:

- **Efficiency:** Performing a batch insert reduces the number of round trips between the client and the server. Instead of making multiple requests, you can send a single request containing all the data entries to be created. This can significantly improve performance and reduce network overhead.
- **Atomicity:** With batch inserts, all the entries are created as a single transaction. This means that either all the entries are successfully inserted, or none of them are. It ensures data consistency and prevents partial or incomplete inserts, maintaining the integrity of the data.
- **Simplified Client-Side Logic:** When creating multiple entries individually, the client application needs to handle error handling, retries, and synchronization. With batch inserts, these concerns are consolidated into a single request, simplifying the client-side logic and making the implementation more straightforward.
- **Network Efficiency:** By sending a batch request, you can take advantage of HTTP pipelining or other optimizations that can reduce the overhead associated with establishing and maintaining separate connections for each individual request.

Overall, batch inserts in a POST REST API provide an efficient and atomic way to create multiple data entries in a single request, streamlining the process and improving performance. Below is an example of a batch POST request:



**Example of a Batch Insert:**

To POST a batch insert, select raw and Text for format. Utilize the POST method to insert several rows into the Fluence client schema table from your planning and forecasting solution, as in the example below.

POST [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/batch/Consolidation\\_Demo](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/batch/Consolidation_Demo)

Body

```
{
  "Account_Name","Audit_Name","CostCenter_Name","DataView_Name","Date_Name","Entity_Name","Intercompany_Name","Movement_Name",
  "Product_Name","Scenario_Name","Amount","Text","Periodic_Amount"
  "6001","Reclassification","000","USA","Value","2023 Dec","210","NoIntercompany","Closing","No product","Budget","30000.00","null","null"
  "6001","Reclassification","000","USA","Value","2023 Dec","220","NoIntercompany","Closing","No product","Budget","50000.00","null","null"
  "6001","Reclassification","000","USA","Value","2023 Dec","510","NoIntercompany","Closing","No product","Budget","60000.00","null","null"
}
```

**NOTE:** The order of the columns must match the order of the columns in the client schema table.

| Option                                                                                  | Description                                                                                                                            |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| POST                                                                                    | REST API Method being used. POST method inserts data into the client schema table.                                                     |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                                               |
| api/                                                                                    | This endpoint indicates the usage of the API.                                                                                          |
| v2305.1/                                                                                | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                         |
| Odata                                                                                   | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API. |
| Batch                                                                                   | This endpoint indicates the loading of batch data.                                                                                     |
| Consolidation_Demo                                                                      | The name of the client schema table within your Fluence environment.                                                                   |
| Body                                                                                    | Inserts a new row within the client schema table based on the metadata intersection within the body.                                   |

**NOTE:** To determine if the POST method worked properly, utilize the GET method using the \$filter query parameter with the intersection of the metadata.

## PUT Method

Utilize the PUT method to update one or more rows to a client schema table. The difference between the POST and PUT methods is that POST inserts a new row into the client schema table, while the PUT method updates an existing row in the client schema table. You must specify a query parameter that will translate to update one or more rows that match the column name.

### Example of a single update:

The example below shows the current intersection within the Consolidation\_Demo client schema table with an Amount of 25000.00. To update that intersection with 45,000.00, we modify the Amount within the body.

PUT [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo?Account\\_Name=6001](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?Account_Name=6001)

Body

```
{
  "Account_Name": "6001",
  "Audit_Name": "Reclassification",
  "CostCenter_Name": "000",
  "Currency_Name": "USA",
  "DataView_Name": "Value",
  "Date_Name": "2023 Dec",
  "Entity_Name": "120",
  "Intercompany_Name": "No Intercompany ",
  "Movement_Name": "Closing",
  "Product_Name": "No product",
  "Scenario_Name": "Budget",
  "Amount": 45000.00,
  "Text": null,
  "Periodic_Amount": null,
}
```

**NOTE:** To determine if the PUT method worked properly, utilize the GET method using the \$filter query parameter with the intersection of the metadata before and after utilizing the PUT method.

## DELETE Method

Use the DELETE method to remove rows from a Fluence client schema table when they are no longer needed or relevant, such as budget data. You must specify a query parameter that will translate to delete one or more rows that match the column name.

**Example of DELETE Method for a single row:**

DELETE [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo)?Account\_Name=6001

Account\_Name= "6001",  
Audit\_Name= "Reclassification",  
CostCenter\_Name= "000",  
Currency\_Name= "USA",  
DataView\_Name= "Value",  
Date\_Name= "2023 Dec",  
Entity\_Name= "120",  
Intercompany\_Name= "No Intercompany ",  
Movement\_Name= "Closing",  
Product\_Name= "No product",  
Scenario\_Name= "Budget",  
Amount= 25000.00,  
Text= null,  
Periodic\_Amount= null

**Example of the DELETE Method to delete all the records from the client schema table while retaining the schema:**

Delete all the records in the client schema table while retaining the schema.

DELETE <https://fluenceapi-prod.fluence.app/api/v2305.1/odata/PlanningActuals>

**NOTE:** To determine if the DELETE method worked properly, utilize the GET method using the \$filter query parameter with the intersection of the metadata before and after utilizing the DELETE method.

## Query Parameters

Query parameters serve as criteria for filtering and sorting, allowing API requests to retrieve specific data. The Fluence RESTAPI utilizes:

- **System query options** - System query options are query string parameters that control the amount and order of the data returned for the URL-identified resource. The names of all system query options are optionally prefixed with a dollar (\$) character.

## System Query Options

For the GET methods, the following rules apply:

- \$apply and \$groupby
- \$count
- \$filter

- \$orderby
- \$select
- \$skip
- \$top

System query options do not apply to the POST, PUT, and DELETE methods or the START and STOP workflows.

## \$apply and \$groupby

Extend the aggregation functionality with the \$apply query parameter. The \$groupby statement groups rows of a column with the same stored value based on a specified function in the statement. Typically, these functions are one of the aggregate functions.

**Example of using \$apply and \$groupby query parameters with the aggregate function:**

In the example below, our goal is to group all the Accounts within the client schema table and calculate the total Amount for that specific account.

GET `https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Planning?$apply=groupby((Account),aggregate(Amount+with+sum+as+Total))`

| Option          | Description                                                              |
|-----------------|--------------------------------------------------------------------------|
| GET             | REST API Method being used. GET retrieves information.                   |
| https://fluence | This is the root URL of your Fluence environment's current REST API URL. |

|                                                                                         |                                                                                                                                        |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| c<br>e<br>a<br>pi<br>-<br>pr<br>o<br>d.<br>fl<br>u<br>e<br>n<br>c<br>e.<br>a<br>p<br>p/ |                                                                                                                                        |
| a<br>pi<br>/                                                                            | This endpoint indicates the usage of the API.                                                                                          |
| v<br>2<br>3<br>0<br>5.<br>1/                                                            | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                         |
| O<br>d<br>at<br>a                                                                       | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API. |
| Pl<br>a<br>n<br>ni<br>n<br>g                                                            | The name of the client schema table within your Fluence environment.                                                                   |
| \$<br>a<br>p<br>pl<br>y=                                                                | \$apply used with \$groupby and the aggregation function.                                                                              |

groupby groups and aggregates all the Accounts in the Planning client schema table and provides a total for the Amounts.

**Results:**

```
[  
  
  {  
  
    "Account": "6401",  
  
    "Total": 22073792.0  
  
  },  
  
  {  
  
    "Account": "6700",  
  
    "Total": 6260634.0  
  
  },  
  
  {  
  
    "Account": "1400",  
  
    "Total": 54903607.0  
  
  }  
  
]
```

## \$count

The \$count system query option allows clients to request a count of the matching resources included with the resources in the response.

### Example of using \$count:

In the example below, the GET method retrieves and counts within the Consolidation\_Demo the number of records for Entity\_Name '120'.

GET [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo?\\$Filter=\(Entity\\_Name eq '120'\)&\\$apply=groupby\(\(Entity\\_Name\),aggregate\(\\$count as Count\)\)](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?$Filter=(Entity_Name%20eq%20'120')&$apply=groupby((Entity_Name),aggregate($count%20as%20Count)))

| Option | Description |
|--------|-------------|
|--------|-------------|

|                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET                                                                                       | REST API Method being used. GET retrieves information.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| https://fluenceapi-prod.fluence.app/                                                      | This is the root URL of your Fluence environment's current REST API URL.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| api/                                                                                      | This endpoint indicates the usage of the API.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v2305.1/                                                                                  | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                                                                                                                                                                                                                                                                                                                                                                           |
| Odata                                                                                     | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API.                                                                                                                                                                                                                                                                                                                                                   |
| Consolidation_Demo                                                                        | The name of the client schema table within your Fluence environment.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Filter=(Entity_Name eq '120')&\$apply=groupby((Entity_Name), aggregate(\$count as Count)) | <p>The \$filter query parameter retrieves only the following columns and fields for the Consolidation_Demo:</p> <ul style="list-style-type: none"> <li>For the column, Entity_Name, any row that equals 120</li> </ul> <p>Extend the aggregation functionality with the \$apply query parameter.</p> <p>The \$groupby statement groups the rows by the Entity_Name column based on the filter, aggregates all Entity_Name rows within the client schema table, and provides a count.</p> |

**Result:**

```
[
  {
```



```
"Entity_Name": "120",  
  
"Count": 205  
  
}  
  
]
```

## \$filter

Clients can use the \$filter system query option to filter a collection of resources addressed by a request URL. The system evaluates the specified expression with the \$filter for each row in the table. Only items that evaluate the expression to be true are included in the response.

### Example of using the \$filter query parameter:

In the example below, the GET method retrieves consolidation data for a specific entity, date, and scenario, which a planning solution utilizes for planning purposes.

GET [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo?\\$filter=\(Entity\\_Name eq '120'\) and \(Date\\_Name eq '2020 Dec'\) and \(Scenario\\_Name eq 'Actual'\)](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?$filter=(Entity_Name eq '120') and (Date_Name eq '2020 Dec') and (Scenario_Name eq 'Actual'))

| Option                                                                                  | Description                                                                                                    |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| GET                                                                                     | REST API Method being used. GET retrieves information.                                                         |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                       |
| api/                                                                                    | This endpoint indicates the usage of the API.                                                                  |
| v2305.1/                                                                                | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL. |

|                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Odata                                                                                           | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API.                                                                                                                                                                                                                                  |
| Consolidation_Demo                                                                              | The name of the client schema table within your Fluence environment.                                                                                                                                                                                                                                                                                                    |
| ?\$filter=(Entity_Name eq '120' ) and (Date_Name eq '2020 Dec') and (Scenario_Name eq 'Actual') | <p>The \$filter query parameter retrieves only the following columns and fields for the Consolidation_Demo:</p> <ul style="list-style-type: none"><li>• For the column, Entity_Name, any row that equals 120 AND</li><li>• For the column, Date_Name, any row that equals 2020 Dec AND</li><li>• For the column, Scenario_Name, any row that equals to Actual</li></ul> |

**Result:**

```
[  
  
  {  
  
    "Account_Name": "2600-240",  
  
    "Audit_Name": "IntercoAdjustment",  
  
    "CostCenter_Name": "000",  
  
    "Currency_Name": "EUR",  
  
    "DataView_Name": "Conversion",  
  
    "Date_Name": "2020 Dec",  
  
    "Entity_Name": "120",  
  
    "Intercompany_Name": "240",  
  
    "Movement_Name": "Variation (Auto)",
```

```
"Product_Name": "No product",

"Scenario_Name": "Actual",

"Amount": 8864.5,

"Text": null,

"Periodic_Amount": null,

},

{

  "Account_Name": "2600-240",

  "Audit_Name": "IntercoAdjustment",

  "CostCenter_Name": "000",

  "Currency_Name": "EUR",

  "DataView_Name": "Conversion",

  "Date_Name": "2020 Dec",

  "Entity_Name": "120",

  "Intercompany_Name": "240",

  "Movement_Name": "CTA_Flow",

  "Product_Name": "No product",

  "Scenario_Name": "Actual",

  "Amount": 456.14,

  "Text": null,

  "Periodic_Amount": null,

},

{

  "Account_Name": "2600-240",

  "Audit_Name": "IntercoAdjustment",

  "CostCenter_Name": "000",
```

```
"Currency_Name": "EUR",  
  
"DataView_Name": "Conversion",  
  
"Date_Name": "2020 Dec",  
  
"Entity_Name": "120",  
  
"Intercompany_Name": "240",  
  
"Movement_Name": "Closing",  
  
"Product_Name": "No product",  
  
"Scenario_Name": "Actual",  
  
"Amount": 9320.64,  
  
"Text": null,  
  
"Periodic_Amount": null,  
  
}  
  
]
```

**NOTE:** More results were returned but are not shown to conserve space in this document.

## \$orderby

Use the \$orderby system query option to specify the order you want to request resources. When utilizing the \$orderby parameter, use 'asc' for ascending and 'desc' for descending.

### Example of \$orderby:

Use the \$orderby to query the fluence client schema table to return the results in ascending or descending order for the data in the Amount field.

GET [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo?\\$orderby=Amount desc](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?$orderby=Amount desc)

| Option | Description |
|--------|-------------|
|--------|-------------|

|                                      |                                                                                                                                              |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| GET                                  | REST API Method being used. GET retrieves information.                                                                                       |
| https://fluenceapi-prod.fluence.app/ | This is the root URL of your Fluence environment's current REST API URL.                                                                     |
| api/                                 | This endpoint indicates the usage of the API.                                                                                                |
| v2305.1/                             | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                               |
| odata                                | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API.       |
| Consolidation_Demo                   | The name of the client schema table within your Fluence tenant.                                                                              |
| ?\$orderby=Amount desc               | The query parameter retrieves all columns and fields for the Consolidation_Demo and orders them in by the Amount fields in descending order. |

**Result:**

```
[
  {
    "Account_Name": "1705",
    "Audit_Name": "OtherManual",
    "CostCenter_Name": "000",
    "Currency_Name": "USD",
    "DataView_Name": "Value",
    "Date_Name": "2020 Dec",
    "Entity_Name": "120",
    "Intercompany_Name": "No Intercompany",
```

```
"Movement_Name": "Increase",

"Product_Name": "No product",

"Scenario_Name": "Actual",

"Amount": 13333332.0,

"Text": null,

"Periodic_Amount": null,

},

{

  "Account_Name": "1705",

  "Audit_Name": "OtherManual",

  "CostCenter_Name": "000",

  "Currency_Name": "USD",

  "DataView_Name": "Value",

  "Date_Name": "2020 Dec",

  "Entity_Name": "120",

  "Intercompany_Name": "Total",

  "Movement_Name": "Increase",

  "Product_Name": "No product",

  "Scenario_Name": "Actual",

  "Amount": 13333332.0,

  "Text": null,

  "Periodic_Amount": null,

}

]
```

**NOTE:** More results were returned but are not shown to conserve space in this document.

# \$select

The \$select system query option retrieves a specific set of columns from the Client schema table. Your Fluence Technologies’ Consolidation system may contain extensive financial data with multiple rows associated with the client schema. By using the \$select parameter, businesses can specify a subset of fields they need in the API response. This reduces network bandwidth and improves performance by retrieving only the required data.

The value of \$select is a comma-separated list of selected items. Each select item is one of the following:

- a star (\*), to include all declared or dynamic columns
- a path to include a column name

**Examples \$select \***

Use the \$select \* to query the Fluence client schema table to analyze the contents. In the example below, we want to explore what dimensions are used in the Consolidation\_Demo, client schema.

**NOTE:** This may retrieve an extensive data set in return.

GET https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\_Demo?\$select \*

| Option                               | Description                                                                                                                            |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| GET                                  | REST API Method being used. GET retrieves information.                                                                                 |
| https://fluenceapi-prod.fluence.app/ | This is the root URL of your Fluence environment's current REST API URL.                                                               |
| api/                                 | This endpoint indicates the usage of the API.                                                                                          |
| v2305.1/                             | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                         |
| odata                                | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence’s REST API. |

|                    |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| Consolidation_Demo | The name of the client schema table within your Fluence tenant.                       |
| ?\$select *        | The query parameter that retrieves all columns and fields for the Consolidation_Demo. |

**Results:**

```
[
  {
    "Account_Name": "2600-240",
    "Audit_Name": "IntercoAdjustment",
    "CostCenter_Name": "000",
    "Currency_Name": "USD",
    "DataView_Name": "Conversion",
    "Date_Name": "2020 Dec",
    "Entity_Name": "140",
    "Intercompany_Name": "Total",
    "Movement_Name": "Closing",
    "Product_Name": "No product",
    "Scenario_Name": "Actual",
    "Amount": 10728.88,
    "Text": null,
    "Periodic_Amount": null,
  },
  {
    "Account_Name": "2600-240",
    "Audit_Name": "IntercoAdjustment",
```



```

    "CostCenter_Name": "000",

    "Currency_Name": "EUR",

    "DataView_Name": "Conversion",

    "Date_Name": "2020 Dec",

    "Entity_Name": "120",

    "Intercompany_Name": "240",

    "Movement_Name": "Variation (Auto)",

    "Product_Name": "No product",

    "Scenario_Name": "Actual",

    "Amount": 8864.5,

    "Text": null,

    "Periodic_Amount": null,

  }
]

```

**NOTE:** More results were returned but are not shown to conserve space in this document.

#### Examples \$select with multiple columns:

In the above example, we retrieved all the columns and rows for the Consolidation\_Demo client schema. In the example below, we need only the Account\_Name, Entity\_Name, CostCenter\_Name, Currency\_Name, Scenario\_Name, Date\_Name, and Amount to populate our planning solution with actuals to assist with budgeting and forecasting for our organization.

**NOTE:** This may retrieve a large data set in return because the request does not contain a \$filter query parameter.

GET [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo?\\$select=Account\\_Name,Entity\\_Name,CostCenter\\_Name,Currency\\_Name,Scenario\\_Name,Date\\_Name,Amount](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?$select=Account_Name,Entity_Name,CostCenter_Name,Currency_Name,Scenario_Name,Date_Name,Amount)

| Option | Description                                            |
|--------|--------------------------------------------------------|
| GET    | REST API Method being used. GET retrieves information. |

|                                                                                                    |                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| https://fluenceapi-prod.fluence.app/                                                               | This is the root URL of your Fluence environment's current REST API URL.                                                                                                                                                                                                                                                  |
| api/                                                                                               | This endpoint indicates the usage of the API.                                                                                                                                                                                                                                                                             |
| v2305.1/                                                                                           | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                                                                                                                                                                                                            |
| Odata                                                                                              | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API.                                                                                                                                                                                    |
| Consolidation_Demo                                                                                 | The name of the client schema table within your Fluence environment.                                                                                                                                                                                                                                                      |
| ?\$select =Account_Name,Entity_Name, CostCenter_Name,Currency_Name, Scenario_Name,Date_Name,Amount | <p>The \$select query parameter retrieves only the following columns and rows for the Consolidation_Demo:</p> <ul style="list-style-type: none"> <li>• Account_Name</li> <li>• Entity_Name</li> <li>• CostCenter_Name</li> <li>• Currency_Name</li> <li>• Scenario_Name</li> <li>• Date_Name</li> <li>• Amount</li> </ul> |

**Results:**

```
[
  {
    "Account_Name": "2600-240",
    "Entity_Name": "140",
    "CostCenter_Name": "000",
    "Currency_Name": "USD",
```

```
"Scenario_Name": "Actual",  
  
"Date_Name": "2020 Dec",  
  
"Amount": 10728.88  
  
},  
  
{  
  
  "Account_Name": "2600-240",  
  
  "Entity_Name": "120",  
  
  "CostCenter_Name": "000",  
  
  "Currency_Name": "EUR",  
  
  "Scenario_Name": "Actual",  
  
  "Date_Name": "2020 Dec",  
  
  "Amount": 8864.5  
  
},  
  
{  
  
  "Account_Name": "2600-240",  
  
  "Entity_Name": "120",  
  
  "CostCenter_Name": "000",  
  
  "Currency_Name": "EUR",  
  
  "Scenario_Name": "Actual",  
  
  "Date_Name": "2020 Dec",  
  
  "Amount": 456.14  
  
}  
  
]
```

**NOTE:** More results were returned but are not shown to conserve space in this document.

# \$skip

The \$skip query option instructs the query to exclude a specified number of items from the collection in the result.

**Example of \$Skip:**

We need to exclude the first 10 rows of the client schema table as they are deemed unnecessary to import into our planning solution.

GET [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo?\\$skip=10](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?$skip=10)

| Option                                                                                  | Description                                                                                                                            |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| GET                                                                                     | REST API Method being used. GET retrieves information.                                                                                 |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                                               |
| api/                                                                                    | This endpoint indicates the usage of the API.                                                                                          |
| v2305.1/                                                                                | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                         |
| Odata                                                                                   | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence’s REST API. |
| Consolidation_Demo                                                                      | The name of the client schema table within your Fluence environment.                                                                   |
| ?\$skip=10                                                                              | The \$skip query parameter skips the first 10 rows in the Fluence client schema table.                                                 |

**Results:**

[  
  
{

```
"Account_Name": "2600-210",  
  
"Audit_Name": "IntercoAdjustment",  
  
"CostCenter_Name": "000",  
  
"Currency_Name": "EUR",  
  
"DataView_Name": "Conversion",  
  
"Date_Name": "2020 Dec",  
  
"Entity_Name": "120",  
  
"Intercompany_Name": "Total",  
  
"Movement_Name": "Variation (Auto)",  
  
"Product_Name": "No product",  
  
"Scenario_Name": "Actual",  
  
"Amount": 8864.5,  
  
"Text": null,  
  
"Periodic_Amount": null,  
  
},
```

```
{  
  
"Account_Name": "2600-210",  
  
"Audit_Name": "IntercoAdjustment",  
  
"CostCenter_Name": "000",  
  
"Currency_Name": "EUR",  
  
"DataView_Name": "Conversion",  
  
"Date_Name": "2020 Dec",  
  
"Entity_Name": "120",  
  
"Intercompany_Name": "Total",  
  
"Movement_Name": "CTA_Flow",  
  
"Product_Name": "No product",
```

```
"Scenario_Name": "Actual",  
  
"Amount": 456.14,  
  
"Text": null,  
  
"Periodic_Amount": null,  
  
}
```

**NOTE:** More results were returned but are not shown to conserve space in this document.

## \$top

The \$top system query option instructs the query to include a specified number of items from the collection in the result.

### Example of \$top:

We only include the first 10 rows of the client schema table in our planning solution.

GET [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation\\_Demo?\\$top=10](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?$top=10)

| Option                                                                                  | Description                                                                                                                            |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| GET                                                                                     | REST API Method being used. GET retrieves information.                                                                                 |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                                               |
| <a href="#">api/</a>                                                                    | This endpoint indicates the usage of the API.                                                                                          |
| <a href="#">v2305.1/</a>                                                                | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                         |
| Odata                                                                                   | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API. |

|                    |                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------|
| Consolidation_Demo | The name of the client schema table within your Fluence environment.                      |
| ?\$top=10          | The \$top query parameter retrieves the first 10 rows in the Fluence client schema table. |

**Result:**

```
[
  {
    "Account_Name": "2600-240",
    "Audit_Name": "IntercoAdjustment",
    "CostCenter_Name": "000",
    "Currency_Name": "USD",
    "DataView_Name": "Conversion",
    "Date_Name": "2020 Dec",
    "Entity_Name": "140",
    "Intercompany_Name": "Total",
    "Movement_Name": "Closing",
    "Product_Name": "No product",
    "Scenario_Name": "Actual",
    "Amount": 10728.88,
    "Text": null,
    "Periodic_Amount": null,
  },
  {
    "Account_Name": "2600-240",
```

```
"Audit_Name": "IntercoAdjustment",  
  
"CostCenter_Name": "000",  
  
"Currency_Name": "EUR",  
  
"DataView_Name": "Conversion",  
  
"Date_Name": "2020 Dec",  
  
"Entity_Name": "120",  
  
"Intercompany_Name": "240",  
  
"Movement_Name": "Variation (Auto)",  
  
"Product_Name": "No product",  
  
"Scenario_Name": "Actual",  
  
"Amount": 8864.5,  
  
"Text": null,  
  
"Periodic_Amount": null,  
  
}  
  
]
```

**NOTE:** More results were returned but are not shown to conserve space in this document.

## Multiple Query Parameters

You can combine different query parameters.

To use multiple query parameters, such as the `$select` and `$filter`, add the `&` after the query parameter as in the example below:



**Example of \$select and \$filter:**

In the example below, we may only want to not only retrieve data for Entity 120, Date 2020 Dec, and Scenario for Actuals but also only return specific columns, such as the Account\_Name, Entity\_Name, CostCenter\_Name, Currency\_Name, Scenario\_Name, Date\_Name, and Amount columns to populate our planning solution.

GET https://fluenceapi-

prod.fluence.app/api/v2305.1/odata/Consolidation\_Demo?\$select=Account\_Name,Entity\_Name,CostCenter\_Name,Currency\_Name,Scenario\_Name,Date\_Name,Amount&\$filter=(Entity\_Name eq '120') and (Date\_Name eq '2020 Dec') and (Scenario\_Name eq 'Actual')

| Option                                                                                                                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET                                                                                                                                                                                               | REST API Method being used. GET retrieves information.                                                                                                                                                                                                                                                                                                                                                                              |
| https://fluenceapi-prod.fluence.app/                                                                                                                                                              | This is the root URL of your Fluence environment's current REST API URL.                                                                                                                                                                                                                                                                                                                                                            |
| api/                                                                                                                                                                                              | This endpoint indicates the usage of the API.                                                                                                                                                                                                                                                                                                                                                                                       |
| v2305.1/                                                                                                                                                                                          | This endpoint indicates the current API Version for Fluence Portal used to create part of the request U                                                                                                                                                                                                                                                                                                                             |
| Odata                                                                                                                                                                                             | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for buildi and consuming Fluence's REST API.                                                                                                                                                                                                                                                                                                |
| Consolidation_Demo                                                                                                                                                                                | The name of the client schema table within your Fluence environment.                                                                                                                                                                                                                                                                                                                                                                |
| ?\$select =Account_Name,Entity_Name, CostCenter_Name,Currency_Name, Scenario_Name,Date_Name,Amount &\$filter=(Entity_Name eq '120') and (Date_Name eq '2020 Dec') and (Scenario_Name eq 'Actual') | <p>The \$select query parameter retrieves only the following columns and fields for the Consolidation_Demo:</p> <ul style="list-style-type: none"> <li>• Account_Name</li> <li>• Entity_Name</li> <li>• CostCenter_Name</li> <li>• Currency_Name</li> <li>• Scenario_Name</li> <li>• Date_Name</li> </ul> <p>Amount</p> <p>The \$filter query parameter that eves only the follo columns and fields for the Consolidation_Demo:</p> |

- For the column, Entity\_Name, any row that equals 120
- For the column, Date\_Name, any row that equals 2020 Dec
- For the column, Scenario\_Name, any row that equals to Actual

**NOTE:** The filter parameter uses the logical operators and to combine multiple filters.

### Results:

```
[
  {
    "Account_Name": "2600-240",
    "Entity_Name": "120",
    "CostCenter_Name": "000",
    "Currency_Name": "EUR",
    "Scenario_Name": "Actual",
    "Date_Name": "2020 Dec",
    "Amount": 8864.5
  },
  {
    "Account_Name": "2600-240",
    "Entity_Name": "120",
    "CostCenter_Name": "000",
    "Currency_Name": "EUR",
    "Scenario_Name": "Actual",
    "Date_Name": "2020 Dec",
```

```
"Amount": 456.14

},

{

  "Account_Name": "2600-240",

  "Entity_Name": "120",

  "CostCenter_Name": "000",

  "Currency_Name": "EUR",

  "Scenario_Name": "Actual",

  "Date_Name": "2020 Dec",

  "Amount": 9320.64

}
```

**NOTE:** More results were returned but are not shown to conserve space in this document.

A client can request specific items from the Fluence environment by combining \$top, \$skip, and \$orderby.

**Example:** We want to retrieve specific columns, get the top 10 rows while skipping the first two rows, and order the Amounts in ascending order.

`https://fluenceapi-prod.fluence.app/api/v2305.1/odata/Consolidation_Demo?$select=Account_Name,Entity_Name,CostCenter_Name,Currency_Name,Scenario_Name,Date_Name,Amount&$top=10&$skip=2&$orderby=Amount desc`

| Option                                            | Description                                                                 |
|---------------------------------------------------|-----------------------------------------------------------------------------|
| GET                                               | REST API Method being used. GET retrieves data from the REST API.           |
| <code>https://fluenceapi-prod.fluence.app/</code> | This is the root URL of your Fluence environment. This is the REST API URL. |
| <code>api/</code>                                 | This endpoint indicates the usage of the REST API.                          |

|                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v2305.1/                                                                                                                                         | This endpoint indicates the current API version of the Fluence Portal used to create part of the demo.                                                                                                                                                                                                                                                                                                                                                                                |
| Odata                                                                                                                                            | This endpoint indicates the OData (Open Data Architecture) which defines the best practice for building Fluence’s REST API.                                                                                                                                                                                                                                                                                                                                                           |
| Consolidation_Demo                                                                                                                               | The name of the client schema table within the demo environment.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| ?\$select =Account_Name,Entity_Name,CostCenter_Name,Currency_Name,Scenario_Name,Date_Name,Amount<br><br>&\$top=10&\$skip=2&\$orderby=Amount desc | <p>The \$select query parameter retrieves the following columns and fields for the Consolidation_Demo table:</p> <ul style="list-style-type: none"><li>• Account_Name</li><li>• Entity_Name</li><li>• CostCenter_Name</li><li>• Currency_Name</li><li>• Scenario_Name</li><li>• Date_Name</li><li>• Amount</li></ul> <p>The \$Top selects the top 10 rows. The \$skip skips the first two rows. The \$orderby parameter orders the data by the Amount column in descending order.</p> |

Result:

```
[
  {
    "Account_Name": "1705",
    "Entity_Name": "120",
    "CostCenter_Name": "000",
    "Currency_Name": "EUR",
    "Scenario_Name": "Actual",
    "Date_Name": "2020 Dec",
    "Amount": 11819334.11
  }
]
```

```
},
```

```
{
```

```
  "Account_Name": "1705",
```

```
  "Entity_Name": "120",
```

```
  "CostCenter_Name": "000",
```

```
  "Currency_Name": "EUR",
```

```
  "Scenario_Name": "Actual",
```

```
  "Date_Name": "2020 Dec",
```

```
  "Amount": 11819334.11
```

```
},
```

```
{
```

```
  "Account_Name": "1705",
```

```
  "Entity_Name": "120",
```

```
  "CostCenter_Name": "000",
```

```
  "Currency_Name": "EUR",
```

```
  "Scenario_Name": "Actual",
```

```
  "Date_Name": "2020 Dec",
```

```
  "Amount": 6213757.55
```

```
},
```

```
{
```

```
  "Account_Name": "1705",
```

```
  "Entity_Name": "120",
```

```
  "CostCenter_Name": "000",
```

```
  "Currency_Name": "USD",
```

```
  "Scenario_Name": "Actual",
```

```
  "Date_Name": "2020 Dec",
```

```
"Amount": 3333333.0

},

{

  "Account_Name": "1705",

  "Entity_Name": "120",

  "CostCenter_Name": "000",

  "Currency_Name": "USD",

  "Scenario_Name": "Actual",

  "Date_Name": "2020 Dec",

  "Amount": 3333333.0

},

{

  "Account_Name": "1705",

  "Entity_Name": "120",

  "CostCenter_Name": "000",

  "Currency_Name": "EUR",

  "Scenario_Name": "Actual",

  "Date_Name": "2020 Dec",

  "Amount": 304090.49

},

{

  "Account_Name": "4000",

  "Entity_Name": "140",

  "CostCenter_Name": "000",

  "Currency_Name": "USD",

  "Scenario_Name": "Actual",
```

```
"Date_Name": "2020 Dec",

"Amount": 56404.75

},

{

  "Account_Name": "4000",

  "Entity_Name": "140",

  "CostCenter_Name": "000",

  "Currency_Name": "USD",

  "Scenario_Name": "Actual",

  "Date_Name": "2020 Dec",

  "Amount": 56404.75

},

{

  "Account_Name": "1100",

  "Entity_Name": "140",

  "CostCenter_Name": "000",

  "Currency_Name": "USD",

  "Scenario_Name": "Actual",

  "Date_Name": "2020 Dec",

  "Amount": 56404.75

},

{

  "Account_Name": "1100",

  "Entity_Name": "140",

  "CostCenter_Name": "000",

  "Currency_Name": "USD",
```

```
"Scenario_Name": "Actual",  
  
"Date_Name": "2020 Dec",  
  
"Amount": 53644.4  
  
}  
  
]
```

## Logical Operators

Fluence's Rest API uses logical operators that evaluate true or false and filter a collection within the resource.

The table below defines the logical operators allowed:

| Operator                        | Description                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Equals (eq)</b>              | The <b>eq</b> operator returns true if the left operand is equal to the right operand. Otherwise, it returns false.                 |
| <b>Not Equals (ne)</b>          | The <b>ne</b> operator returns true if the left operand is not equal to the right operand. Otherwise, it returns false.             |
| <b>Great Than (gt)</b>          | The <b>gt</b> operator returns true if the left operand is greater than the right operand. Otherwise, it returns false.             |
| <b>Great Than or Equal (ge)</b> | The <b>ge</b> operator returns true if the left operand is greater than or equal to the right operand. Otherwise, it returns false. |
| <b>Less Than (lt)</b>           | The <b>lt</b> operator returns true if the left operand is less than the right operand. Otherwise, it returns false.                |
| <b>Less Than or Equal (le)</b>  | The <b>le</b> operator returns true if the left operand is less than or equal to the right operand. Otherwise, it returns false.    |



|                   |                                                                                                                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>And</b>        | The <b>and</b> operator returns true if both the left and right operands evaluate to true. Otherwise, it returns false.                                                                                                                                                                                      |
| <b>Or</b>         | The <b>or</b> operator returns false if both the left and right operands both evaluate to false. Otherwise, it returns true.                                                                                                                                                                                 |
| <b>Not</b>        | The <b>not</b> operator returns true if the operand returns false. Otherwise, it returns false.                                                                                                                                                                                                              |
| <b>Has</b>        | The <b>has</b> operator returns true if the right operand is an enumeration value whose flag(s) are set on the left operand.                                                                                                                                                                                 |
| <b>In</b>         | The <b>in</b> operator returns true if the left operand is a member of the right operand. The right operand <b>MUST</b> be either a comma-separated list of primitive values enclosed in parentheses or a single expression that resolves to a collection.                                                   |
| <b>Endswith</b>   | The <b>endswith</b> function with string parameter values returns true if the first string ends with the second string. Otherwise, it returns false. String comparison is case-sensitive. Case-insensitive comparison can be achieved in combination with <code>tolower</code> or <code>toupper</code> .     |
| <b>Startswith</b> | The <b>startswith</b> function with string parameter values returns true if the first string starts with the second string. Otherwise, it returns false. String comparison is case-sensitive. Case-insensitive comparison can be achieved in combination with <code>tolower</code> or <code>toupper</code> . |

## Equals (eq)

The **eq** operator returns true if the left operand is equal to the right operand. Otherwise, it returns false.

### Example of eq:

If you would like to return all of the balance sheet statement amounts for just a particular date, such as 2020 Dec, you would write the following:

GET [https://fluenceapi-prod.fluence.app/api/v2305.1/odata/BalSheet?\\$filter=Date eq '2020 Dec'](https://fluenceapi-prod.fluence.app/api/v2305.1/odata/BalSheet?$filter=Date eq '2020 Dec')

| Option                                                                                  | Description                                                                                                                            |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| GET                                                                                     | REST API Method being used. GET retrieves information.                                                                                 |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                                               |
| api/                                                                                    | This endpoint indicates the usage of the API.                                                                                          |
| v23085.1/                                                                               | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL.                         |
| Odata                                                                                   | This endpoint indicates the OData (Open Data Protocol), which defines the best practice for building and consuming Fluence's REST API. |
| BalSheet                                                                                | The name of the client schema table within your Fluence environment.                                                                   |
| ?\$filter=Date eq '2020 Dec'                                                            | The query parameter filters the Date column and retrieves only the rows equaling '2020 Dec' for the BalSheet.                          |

### Results:

[

```
{  
  
  "Account": "Balance Sheet Accounts",  
  
  "Date": "2020 Dec",  
  
  "Audit": "Contribution",  
  
  "CostCenter": "Total CostCenter",  
  
  "Currency": "USD",  
  
  "DataView": "Value",  
  
  "Entity": "Total_Group",  
  
  "Intercompany": "No Intercompany",  
  
  "Movement": "Closing",  
  
  "Product": "No product",  
  
  "Scenario": "Actual",  
  
  "Amount": "-3326323"  
},
```

```
{  
  
  "Account": "Balance Sheet Accounts",  
  
  "Date": "2020 Dec",  
  
  "Audit": "Contribution",  
  
  "CostCenter": "Total CostCenter",  
  
  "Currency": "USD",  
  
  "DataView": "Value",  
  
  "Entity": "USA",  
  
  "Intercompany": "No Intercompany",  
  
  "Movement": "Closing",  
  
  "Product": "No product",  
  
  "Scenario": "Actual",
```

```
"Amount": "-3326323"
}
]
```

**NOTE:** More results were returned but are not shown to conserve space in this document.

## Operator Parameters

When utilizing the \$filter and \$orderby query parameters, use the following operator precedence for supported operators. Equal precedence is given to operators in the same category.

| Group      | Operator | Description           |
|------------|----------|-----------------------|
| Grouping   | ( )      | Precedence grouping   |
| Unary      | -        | Negation              |
|            | not      | Logical Negation      |
|            | cast( )  | Type Casting          |
| Relational | gt       | Greater Than          |
|            | ge       | Greater than or Equal |
|            | lt       | Less Than             |
|            | le       | Less than or Equal    |
|            | isof     | Type Testing          |
| Equality   | eq       | Equal                 |
|            | ne       | Not Equal             |

|                 |     |             |
|-----------------|-----|-------------|
| Conditional AND | and | Logical And |
| Conditional OR  | or  | Logical Or  |

## About Syncing Dimension Information with External Applications Using Fluence's REST API

Synchronize dimension information with an external application, like a planning solution, by accessing Fluence's REST API. In this scenario, Fluence acts as the source of dimensional information.

**Use Case:** Dimension Data Retrieval for external solutions, such as a Business Intelligence or Planning solution.

### Purpose:

From Fluence, through its REST API, retrieve a list of dimensions, including parent-child relationships and attributes. This data will populate external solutions, such as business intelligence and planning solutions. The goal is to maintain consistent and synchronized metadata across both systems.

### Benefits:

By obtaining dimension details via the API, the planning software gains access to the financial structure and hierarchy utilized for actuals data in the Financial Consolidation solution. This enhances management reporting, planning, and forecasting accuracy and reliability within the planning software. The predefined dimensions and relationships facilitate proper data alignment and analysis, eliminating manual upkeep. This approach minimizes the risk of structural disparities or incongruities between the two systems, ultimately simplifying system administration.

**API Usage:**

Users can utilize our REST API in JSON format to:

Retrieve a list of:

- Dimensions and their properties within the entire tenant.
- Dimensions and their associated properties within a specific model.
- Members and their associated properties within a specific dimension.

**API Features:**

Supports the GET method.

The GET request retrieves information from the Fluence solution. By specifying the relevant parameters in the GET request, users can retrieve consolidated financial information such as a list of dimensions and properties within the entire system, dimensions and properties based on a model, and members with properties for a specific dimension. This integration facilitates data sharing, synchronization, and real-time updates across different systems, enhancing overall operational efficiency and accuracy of financial information.

Provides vital information, including member name, caption, dimension attributes, and parent-child relationships.

Member keys can be included in queries, although they might not be relevant outside Fluence. Within Fluence, when you need to update or delete a specific member, using its key ensures that you're targeting the exact item you intend to modify or remove.

## Retrieving a List of Dimensions and Properties Within a Tenant

Retrieving a comprehensive list of dimensions and their properties from Fluence through our REST API serves the fundamental purpose of seamlessly integrating critical financial data into external applications. By leveraging the API to access this information, organizations can establish a dynamic connection between their Fluence solution and various other software systems, such as reporting tools or planning applications. This integration ensures that the most up-to-date dimension details are readily available for analysis, budgeting, and

decision-making across platforms. By employing the REST API, organizations enable efficient data sharing, minimize manual data entry, and maintain data consistency between systems, thereby optimizing their financial processes and enhancing the accuracy of their strategic initiatives.

Use the GET method to fetch the complete list of dimensions in your Fluence Solution.

**Example of retrieving the list of dimensions within a tenant**

GET <https://fluenceapi-prod.fluence.app/api/v2305.1/Dimensions>

| Option                                                                                  | Description                                                                                                    |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| GET                                                                                     | REST API Method being used. GET retrieves information.                                                         |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                       |
| <a href="#">api/</a>                                                                    | This endpoint indicates the usage of the API.                                                                  |
| <a href="#">v2305.1/</a>                                                                | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL. |
| <a href="#">Dimensions</a>                                                              | Retrieves a list of all the dimensions and their properties from your entire Fluence solution.                 |

**Results of the GET Method above:**

Below is a sample of the results from the GET method above:

Response:

json

```
[
{
  "dimensionName": "Account",
  "dimensionType": "Account",
  "translations": "fr",
  "relatedDimension": null,
  "properties": [
    {
      "propertyName": "Classification",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": true,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "Consolidation",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "Conversion",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "CopyOpening",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
```



```
"typeInfo": null
},
{
  "propertyName": "DebitCredit",
  "propertyType": "string",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "Journal",
  "propertyType": "Bit",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "MovementClass",
  "propertyType": "string",
  "standardProperty": false,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "SysName",
  "propertyType": "List",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "TimeBalance",
  "propertyType": "string",
  "standardProperty": true,
  "reportingProperty": false,
```

```
"readonlyProperty": false,
"processingStatus": 0,
"typeInfo": null
},
{
  "propertyName": "Variation",
  "propertyType": "Bit",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
}
],
"processingStatus": null
},
]
```

**NOTE:** More results were returned but are removed for brevity.

## Retrieving a List of Dimensions and Properties Within a Specific Model

The purpose of retrieving a comprehensive list of dimensions and their properties within a specific model from your Fluence solution using our REST API is to facilitate focused and efficient financial analysis and management. This process allows organizations to extract dimension-related data tailored to a specific model, enabling more targeted insights and decision-making. By employing the REST API for this purpose, organizations can seamlessly integrate these dimension details into their chosen applications, such as reporting or planning tools, ensuring that the analysis is contextually relevant and aligned with the designated financial model. This integration streamlines data accessibility, enhances accuracy, and empowers organizations to derive actionable intelligence while maintaining consistency across their financial processes.

Use the GET method to fetch the complete list of dimensions specific to a model in your Fluence Solution.

**Example of retrieving the list of dimensions within a specific model**

GET <https://fluenceapi-prod.fluence.app/api/v2305.1/models/Consolidations/Dimensions>

| Option                                                                                  | Description                                                                                                    |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| GET                                                                                     | REST API Method being used. GET retrieves information.                                                         |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                       |
| <a href="#">api/</a>                                                                    | This endpoint indicates the usage of the API.                                                                  |
| <a href="#">v2305.1/</a>                                                                | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL. |
| <a href="#">models</a>                                                                  | This endpoint indicates the retrieval of a list of dimensions specific to a model.                             |
| <a href="#">Consolidations</a>                                                          | The name of the model being utilized.                                                                          |
| <a href="#">Dimensions</a>                                                              | Retrieves a list of all the dimensions and their properties from your entire Fluence solution.                 |

**Results of the GET Method above:**

Below is a sample of the results from the GET method above:

Response:

json

```
[
{
  "dimensionName": "Account",
  "dimensionType": "Account",
  "translations": "fr",
  "relatedDimension": null,
  "properties": [
    {
      "propertyName": "Classification",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": true,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "Consolidation",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "Conversion",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "CopyOpening",
      "propertyType": "string",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
  ],
}
```

```
{
  "propertyName": "DebitCredit",
  "propertyType": "string",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "Journal",
  "propertyType": "Bit",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "MovementClass",
  "propertyType": "string",
  "standardProperty": false,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "SysName",
  "propertyType": "List",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "TimeBalance",
  "propertyType": "string",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
```

```
"typeInfo": null
},
{
  "propertyName": "Variation",
  "propertyType": "Bit",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
}
],
"processingStatus": null
},
{
  "dimensionName": "Audit",
  "dimensionType": "Audit",
  "translations": null,
  "relatedDimension": null,
  "properties": [
    {
      "propertyName": "AuditType",
      "propertyType": "List",
      "standardProperty": false,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "Calculated",
      "propertyType": "Bit",
      "standardProperty": false,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "Consolidated",
      "propertyType": "Bit",
      "standardProperty": true,
      "reportingProperty": false,
```

```
"readonlyProperty": false,
"processingStatus": 0,
"typeInfo": null
},
{
  "propertyName": "Converted",
  "propertyType": "Bit",
  "standardProperty": true,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "CopyNetIncome",
  "propertyType": "bool",
  "standardProperty": false,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
},
{
  "propertyName": "PreviousLevel",
  "propertyType": "Dimension",
  "standardProperty": false,
  "reportingProperty": false,
  "readonlyProperty": false,
  "processingStatus": 0,
  "typeInfo": null
}
],
"processingStatus": null
},
{
  "dimensionName": "CostCenter",
  "dimensionType": "Other",
  "translations": "",
  "relatedDimension": null,
  "properties": null,
  "processingStatus": null
},
{
```

```
"dimensionName": "Currency",
"dimensionType": "Currency",
"translations": null,
"relatedDimension": null,
"properties": [
{
"propertyName": "BaseCurrency",
"propertyType": "bit",
"standardProperty": true,
"reportingProperty": false,
"readonlyProperty": false,
"processingStatus": 0,
"typeInfo": null
},
{
"propertyName": "RateOperator",
"propertyType": "List",
"standardProperty": true,
"reportingProperty": false,
"readonlyProperty": false,
"processingStatus": 0,
"typeInfo": null
},
{
"propertyName": "Reporting",
"propertyType": "Bit",
"standardProperty": true,
"reportingProperty": false,
"readonlyProperty": false,
"processingStatus": 0,
"typeInfo": null
},
{
"propertyName": "Triangulate",
"propertyType": "bit",
"standardProperty": true,
"reportingProperty": false,
"readonlyProperty": false,
"processingStatus": 0,
"typeInfo": null
}
],
"processingStatus": null
```



```
},
{
  "dimensionName": "DataView",
  "dimensionType": "DataView",
  "translations": null,
  "relatedDimension": null,
  "properties": [
    {
      "propertyName": "Consolidated",
      "propertyType": "Bit",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "Converted",
      "propertyType": "Bit",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    },
    {
      "propertyName": "DataviewType",
      "propertyType": "List",
      "standardProperty": true,
      "reportingProperty": false,
      "readonlyProperty": false,
      "processingStatus": 0,
      "typeInfo": null
    }
  ],
  "processingStatus": null
},
]
```

**NOTE:** More results were returned but are removed for brevity.

# Retrieving a List of Dimension Members and Properties

Retrieving dimension members and their properties from your Fluence solution through our REST API serves the crucial purpose of accessing and integrating granular financial data. This process enables organizations to extract detailed information about individual dimension members, such as accounts, entities, or product categories, along with their associated attributes. By utilizing the REST API for this purpose, organizations can seamlessly incorporate this data into their analysis, reporting, and planning tools, ensuring a comprehensive understanding of their financial landscape.

Use the GET method to fetch the complete list of dimension members and their properties of a specific dimension.

## Batch requests/Server Side paging

Batch requests support server-side paging allowing you to set the query page and the number of rows to retrieve for each page. The maximum paging size is 2500 rows. For the '?Page' field, you can specify the desired query page to retrieve, while the '?PageSize' field in the query specifies how many rows to fetch for that page. The '?Skip' parameter in the query string determines the starting point for rows in server-side paging.

The "?Page" parameter starts at 1

Furthermore, the count of dimension members is also returned.

The 'TotalRows' field, present in the response body, holds the overall row count for the dimension within the system. Similarly, the 'CurrentPage' field in response bodies indicates the current page, and 'TotalPages' represents the total number of pages.

To control the presentation, you can use the 'Format' header key/value. Setting it to 'flat' results in a flattened list of dimension member and property fields. If no format is specified, the default value 'default' is applied. Using the default format, the members and properties are presented in a nested array.

In the example below, the request is configured to fetch the first page of data, containing up to 500 items, from the specified endpoint related to the dimension called Measure within the Fluence solution.

**Example of retrieving the list of dimension members and their properties:**

GET <https://fluenceapi-prod.fluence.app/api/v2305.1/metadata/Dimensions/Measure?Page=1&PageSize=500>

| Option                                                                                  | Description                                                                                                    |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| GET                                                                                     | REST API Method being used. GET retrieves information.                                                         |
| <a href="https://fluenceapi-prod.fluence.app/">https://fluenceapi-prod.fluence.app/</a> | This is the root URL of your Fluence environment's current REST API URL.                                       |
| api/                                                                                    | This endpoint indicates the usage of the API.                                                                  |
| v2305.1/                                                                                | This endpoint indicates the current API Version for the Fluence Portal used to create part of the request URL. |
| metadata                                                                                | This endpoint indicates the retrieval of a list of dimension members and their properties.                     |
| Dimensions                                                                              | Retrieves a list of all the dimensions and their properties from your entire Fluence solution.                 |
| Measure                                                                                 | The name of the dimension used to retrieve its members and associated properties                               |
| ?Page=1&PageSize=500                                                                    | Query parameters serve as criteria to fetch the first page of data, containing up to 500 items.                |

**NOTE:** When retrieving the members and their associated properties, you have the ability to choose the format. By default, the properties are presented in a nested array. Alternatively,

you can opt for the flat format, which displays the associated properties in a list.

### Results of the GET Method above:

Below is a sample of the results from the GET method above:

## Response format set to default:

json

```
{
  "dimensionMembers": [
    {
      "memberName": "Amount",
      "memberTag": "#Amount#",
      "memberCaption": "Amount",
      "ancestors": "Measure|Amount",
      "translations": [],
      "parentMemberName": "",
      "sortOrder": 1,
      "isLeaf": true,
      "operator": null,
      "startDate": null,
      "endDate": null,
      "memberStorage": "Stored",
      "properties": [
        {
          "propertyName": "DataType",
          "propertyValue": "float"
        },
        {
          "propertyName": "TimeBalance",
          "propertyValue": "Sum"
        }
      ]
    },
    {
      "memberName": "Text",
```

```
"memberTag": null,
"memberCaption": "Text",
"ancestors": "Measure|Text",
"translations": [],
"parentMemberName": "",
"sortOrder": 2,
"isLeaf": true,
"operator": null,
"startDate": null,
"endDate": null,
"memberStorage": "Stored",
"properties": [
  {
    "propertyName": "DataType",
    "propertyValue": "String"
  },
  {
    "propertyName": "TimeBalance",
    "propertyValue": "Last"
  }
]
},
{
  "memberName": "Periodic_Amount",
  "memberTag": null,
  "memberCaption": "Periodic_Amount",
  "ancestors": "Measure|Periodic_Amount",
  "translations": [],
  "parentMemberName": "",
  "sortOrder": 3,
  "isLeaf": true,
  "operator": null,
  "startDate": null,
  "endDate": null,
  "memberStorage": "Stored",
  "properties": [
    {
      "propertyName": "DataType",
      "propertyValue": "float"
    },
    {
      "propertyName": "TimeBalance",
      "propertyValue": "Last"
    }
  ]
}
```

```
    }
  ]
},
{
  "memberName": "Test_Gali",
  "memberTag": "",
  "memberCaption": "Test_Gali",
  "ancestors": "Measure|Test_Gali",
  "translations": [],
  "parentMemberName": "",
  "sortOrder": 4,
  "isLeaf": true,
  "operator": null,
  "startDate": null,
  "endDate": null,
  "memberStorage": "Stored",
  "properties": [
    {
      "propertyName": "DataType",
      "propertyValue": "float"
    },
    {
      "propertyName": "TimeBalance",
      "propertyValue": "Sum"
    }
  ]
}
],
"totalRows": 4,
"currentPage": 1,
"totalPages": 1
}
```

## Response format set to flat:

json

```
{
  "dimensionMembers": [
    {
      "MemberName": "Amount",
      "MemberTag": "#Amount#",
      "MemberCaption": "Amount",
      "Ancestors": "Measure|Amount",
      "ParentMemberName": "",
      "SortOrder": 1,
      "IsLeaf": true,
      "Operator": null,
      "StartDate": null,
      "EndDate": null,
      "MemberStorage": "Stored",
      "p_DataType": "float",
      "p_TimeBalance": "Sum"
    },
    {
      "MemberName": "Text",
      "MemberTag": null,
      "MemberCaption": "Text",
      "Ancestors": "Measure|Text",
      "ParentMemberName": "",
      "SortOrder": 2,
      "IsLeaf": true,
      "Operator": null,
      "StartDate": null,
      "EndDate": null,
      "MemberStorage": "Stored",
      "p_DataType": "String",
      "p_TimeBalance": "Last"
    },
    {
      "MemberName": "Periodic_Amount",
      "MemberTag": null,
      "MemberCaption": "Periodic_Amount",
      "Ancestors": "Measure|Periodic_Amount",
      "ParentMemberName": "",
      "SortOrder": 3,
      "IsLeaf": true,
      "Operator": null,
      "StartDate": null,
      "EndDate": null,
    }
  ]
}
```

```
"MemberStorage": "Stored",
"p_DataType": "float",
"p_TimeBalance": "Last"
},
{
"MemberName": "Test_Gali",
"MemberTag": "",
"MemberCaption": "Test_Gali",
"Ancestors": "Measure|Test_Gali",
"ParentMemberName": "",
"SortOrder": 4,
"IsLeaf": true,
"Operator": null,
"StartDate": null,
"EndDate": null,
"MemberStorage": "Stored",
"p_DataType": "float",
"p_TimeBalance": "Sum"
}
],
"totalRows": 4,
"currentPage": 1,
"totalPages": 1
}
```

## START Workflow Process

Utilize START to request that the Fluence server starts a workflow process within the appropriate folder(s) within the **Explorer** module.

Utilize the POST method with the start/. After the REST API version, use process/ and then start/. After the start/, type the name of the folder path separated with the "/". Lastly, type the name of the workflow being started.

The content type needed is JSON. Within the request's body, type the parameters for the Start task within the workflow process or type "[ ]".

**NOTE:** Special characters that are used in the request, such as "?", "&", and "/", are not recommended to be used in the name of the **Explorer** module folders or workflow



processes.

#### Example of START request for a workflow process:

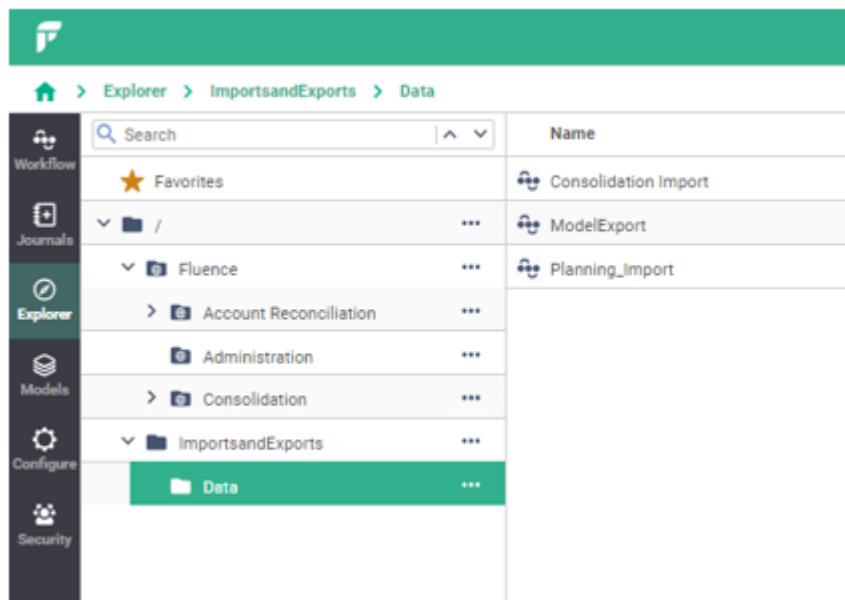
We need to start the ModelExport workflow, which exports the Actuals data from the Consolidation model, creates a .csv file, and stores it in the same Data folder within the *Explorer* module.

POST <https://fluenceapi-prod.fluence.app/api/v2305.1/process/start/ImportsandExports/Data/ModelExport>

Body

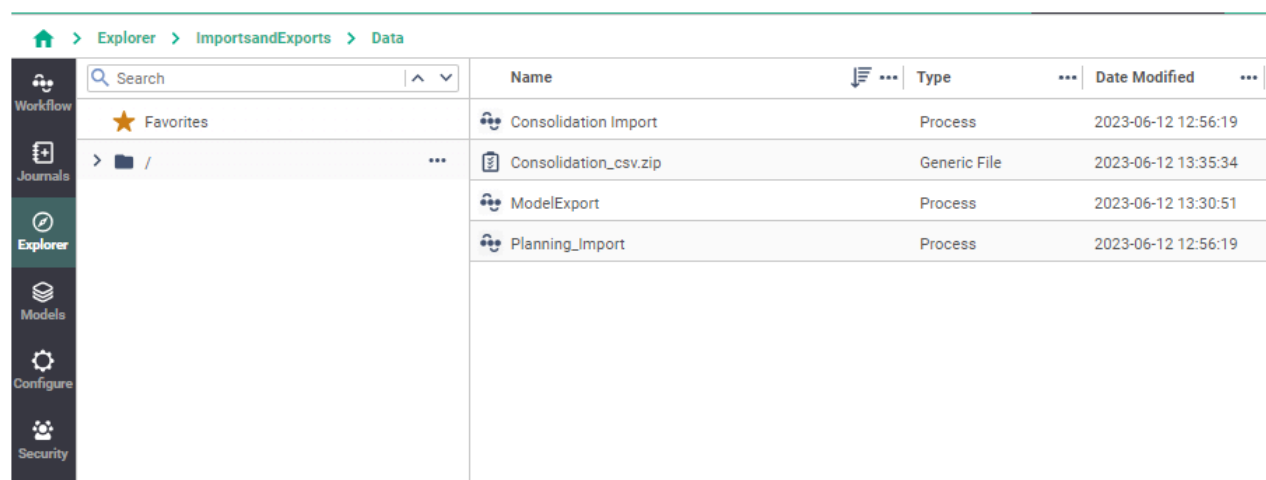
```
[]
```

#### Before starting the workflow process:



**NOTE:** There are only Fluence processes in this folder.

#### After the workflow process has been completed:



## STATE of a Workflow Process

Utilize STATE to request that the Fluence server the status of a workflow process within the appropriate folder(s) within the **Explorer** module.

Utilize the GET method with the state/. After the REST API version, use process/ and then state/. After the state/, type the name of the folder path separated with the "/". Lastly, type the name of the workflow being started.

The content type needed is JSON.

When you use STATE, the workflow process will return the run ID.

**NOTE:** Special characters that are used in the request, such as "?", "&", and "/", are not recommended to be used in the name of the **Explorer** module folders or workflow processes.

### Example of STATE request for a workflow process:

We need to check the state of the ModelExport workflow, which exports the Actuals data from the Consolidation model, creates a .csv file, and stores it in the same Data folder within the *Explorer* module.

POST <https://fluenceapi-prod.fluence.app/api/v2305.1/process/state/ImportsandExports/Data/ModelExport>

Body

```
"00000000-0000-0000-0000-000000000000"
```

## STOP a Workflow Process

Utilize STOP to request that the Fluence server stops a Workflow process within the appropriate folder(s) within the **Explorer** module.

Utilize the POST method with the stop. After the REST API version, use /process and then /stop. After the /stop, type the folder path name separated by the "/". Lastly, type the name of the workflow being started.

**NOTE:** Special characters that are used in the request, such as "?", "&", and "/", are not recommended to be used in the name of the **Explorer** module folders or workflow processes.

**Example of STOP request for a workflow process:**

We need to start the ModelExport workflow, which exports the Actuals data from the Consolidation model, creates a .csv file, and stores it in the same Data folder within the *Explorer* module.

POST [https://fluenceapi-prod.fluence.app/api/v2305.1/process/stop/Data and Metadata Imports and Exports/Data/ModelExport](https://fluenceapi-prod.fluence.app/api/v2305.1/process/stop/Data%20and%20Metadata%20Imports%20and%20Exports/Data/ModelExport)

## About Security Endpoints

Fluence's REST API provides powerful functionality for managing users and security information. Use this feature to automate user management tasks, perform bulk operations, and quickly update security settings, ensuring an efficient and secure environment. You can use the API to perform a variety of tasks:

- **Retrieve User Information:** You can get a list of users and their security details, including user ID, user name, full name, enabled/disabled status, email, and authorization roles.
- **Add or Delete Users:** Add new users or delete existing ones.
- **Update User Information:** Update user details such as their email or their enabled/disabled status from a tenant.
- **Manage User Roles:** Retrieve a list of user roles, assign users to specific roles, or remove users from roles.

## Retrieve User Information

With Fluence's REST API, you can retrieve a list of users and their security information, such as user ID, user name, enabled/disabled status, email, and assigned authorization roles. Use this feature to quickly access detailed user and security information for auditing, troubleshooting, or integrating with other systems.

Use the GET method. You must have "users" after the version in the URL. Ensure the content type is set to 'application/json' and specify the tenant for which you want to retrieve user information.

**Example:**

GET <https://fluenceapi-prod.fluence.app/api/v2305.1/users>

**Results:**

[

```
{
  "userId": 1893,
  "username": "asmith@fluence.app",
  "fullName": null,
  "isdisabled": false,
  "email": "asmith@abccompany.com",
  "roles": [
    "Admin"
  ],
},
{
  "userId": 2567,
  "username": "ssimpson@fluence.app",
  "fullname": "Sally Simpson",
  "isdisabled": false,
  "email": "ssimpson@abccompany.com",
  "roles": [
    "Admin"
  ]
}
```

## Retrieve information for an individual user

To retrieve information for an individual user, add the user name after the "users" in the URL.

### Example:

GET <https://fluenceapi-prod.fluence.app/api/v2305.1/users/ssimpson@fluence.app>

### Results:

```
[
  {
    "userId": 2567,
    "username": "ssimpson@fluence.app",
    "fullname": "Sally Simpson",
```

```
"isdisabled": false,  
"email": "ssimpson@abccompany.com",  
"roles": [  
  "Admin"  
]
```

## Adding Users

With Fluence's REST API, you can add a user to a tenant using the POST method, streamlining user management within a tenant.

You must have "users" after the version in the URL. Ensure that the content type is set to 'application/json' and specify the tenant for which you want to add users. Include the user data within the request body.

### Example:

POST <https://fluenceapi-prod.fluence.app/api/v2305.1/users>

### Body

```
{  
  "username": "asmith@fluence.app",  
  "fullName": null,  
  "isdisabled": false,  
  "email": "asmith@abccompany.com",  
  "roles": [  
    "Admin"  
  ]  
}
```

**Note:** You can enable or disable users from the tenant within the request body.

## Updating User Information

With Fluence's REST API, update a user's security information, such as their email, or enable/disable them from the tenant using the PUT method. Use this feature to update user

details or change their access status, for example, providing user information remains current and accurate within the tenant.

You must have "users" after the version in the URL. Ensure that the content type is set to 'application/json' and specify the tenant for which you want to update the user. Include the updated user data within the request body as shown below:

**Example:**

PUT <https://fluenceapi-prod.fluence.app/api/v2305.1/users>

**Body**

```
[
{
  "username": "asmith@fluence.app",
  "fullName": null,
  "isdisabled": false,
  "email": "asmith@abccompany.com",
  "roles": [
    "Admin"
  ]
}
```

## Deleting Users

Using Fluence's REST API, use the DELETE method to delete a single user or perform a bulk delete of users from a tenant. This feature allows you to remove users who no longer need tenant access, making sure your user base has security. It's useful for cleaning up inactive accounts or managing user turnover.

You must have "users" after the version in the URL. Ensure that the content type is set to 'application/json' and specify the tenant for which you want to delete the users. Include the updated user data within the request body. If users are deleted in bulk, separate them with a comma in the request body.

**Example:**

DELETE <https://fluenceapi-prod.fluence.app/api/v2305.1/users>

## Body

```
{
  "username": "asmith@fluence.app",
  "fullName": null,
  "isdisabled": false,
  "email": "asmith@abccompany.com",
  "roles": [
    "Admin"
  ]
}
```

## Retrieving User Roles

You can retrieve a list of roles assigned to a specific user using the GET method with Fluence's REST API. This feature helps audit user permissions and manage access control.

You must have "user" after the version in the URL. Ensure that the content type is set to 'application/json' and specify the tenant for which you want to retrieve a user's assigned roles. Use the "user" endpoint after the version in the URL, followed by the user name, and then "/roles".

### Example:

GET <https://fluenceapi-prod.fluence.app/api/v2305.1/user/asmith@fluence.app/roles>

### Results:

```
{
  "userId": 1893,
  "username": "asmith@fluence.app",
  "fullName": null,
  "isdisabled": false,
  "email": "asmith@abccompany.com",
  "roles": [
    "Admin"
  ]
}
```

## Removing Roles from Users

Using the DELETE method, you can remove users' roles with Fluence's REST API. This feature manages user permissions by removing unnecessary roles. It makes sure users have only the access they need while maintaining security and clarity in role assignments.

Make sure that the content type is set to 'application/json' and specify the tenant for which you want to remove the roles from users. Include the user roles within the request body. If multiple roles need to be removed from a user, separate the existing roles with a comma.

**Example:**

DELETE <https://fluenceapi-prod.fluence.app/api/v2305.1/user/asmith@fluence.app/roles>

**Body**

"Reviewer", "Contributor"

## Assign Authentication Roles to Users

You can assign user roles using the POST method with Fluence's REST API. Use this feature to manage user permissions efficiently. Assigning specific roles makes sure that users have appropriate access levels and responsibilities.

Specify the user name you want to assign after "user" in the URL followed by "roles". Guarantee that the content type is set to 'application/json' and specify the tenant for which you want to assign the role to a user. Include the name of an existing role, such as "Reviewer" or "Contributor" within the request body. To assign multiple roles, separate the roles with a comma.

**Example:**

```
curl -X POST https://fluenceapi-prod.fluence.app/api/v2305.1/user/asmith@fluence.app/roles \
  -H 'Content-Type: application/json' \
  -d '{"roles": ["Reviewer", "Contributor"]}'
```

Follow Fluence

Visit [fluencetech.com](https://fluencetech.com)

© 2022 Fluence All Rights Reserved